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Math 31 - Differential Equations
Section 7.2.2

38. $y' - y = 2\cos(5t) \mid y(0) = 0$

First, let us take the Laplace Transform of each side:

$$\mathcal{L}\{y' - y\} = \mathcal{L}\{2\cos(5t)\} \longrightarrow sY(s) - y(0) - Y(s) = \frac{2s}{s^2+25}$$

Factoring and adding over $y(0)$ we get:

$$Y(s)(s-1) = \frac{2s}{s^2+25} + 0$$

Dividing by $(s-1)$ results in:

$$Y(s) = \frac{2s}{(s-1)(s^2+25)}$$

Here we must take the inverse Laplace and thus need to make use of PFD. $(s-1)$ is a reducible linear term, however (s^2+25) does not have any real zeroes therefore we must make use of a linear term in our PFD.

$$\frac{2s}{(s-1)(s^2+25)} = \frac{As+B}{(s^2+25)} + \frac{C}{s-1}$$

Simplifying further:

$$\frac{2s}{(s-1)(s^2+25)} = \frac{As^2-As+Bs-B+Cs^2+25C}{(s-1)(s^2+25)}$$

Comparing coefficients we get:

$$A + C = 0$$
$$-A + B = 2$$
$$-B + 25C = 0$$

Setting up our augment matrix: $\begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & -1 & 25 & 0 \end{pmatrix} \longrightarrow$ RREF: $\begin{pmatrix} 1 & 0 & 0 & -\frac{1}{13} \\ 0 & 1 & 0 & \frac{25}{13} \\ 0 & 0 & 1 & \frac{1}{13} \end{pmatrix}$

Giving us:

$$A = -\frac{1}{13}, B = \frac{25}{13}, C = \frac{1}{13}$$

Rewriting our original fraction in its decomposed form gives us:

$$\frac{-\frac{1}{13}s+\frac{25}{13}}{(s^2+25)} + \frac{\frac{1}{13}}{(s-1)}$$

Simplyfing further we get:

$$Y(s) = \frac{25-s}{13(s^2+25)} + \frac{1}{13(s-1)}$$

To get the solution for our DE, we will now take the inverse Laplace:

$$y = \mathcal{L}^{-1}\left\{\frac{-s+25}{13(s^2+25)}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{13(s-1)}\right\}$$

Splitting the transformation results in:

$$y = -\frac{1}{13}\mathcal{L}^{-1}\left\{\frac{s}{s^2+25}\right\} + \frac{25}{13}\mathcal{L}^{-1}\left\{\frac{1}{s^2+25}\right\} + \frac{1}{13}\mathcal{L}^{-1}\left\{\frac{1}{(s-1)}\right\}$$

Getting the inverse Laplace transform into form:

$$y = -\frac{1}{13}\mathcal{L}^{-1}\left\{\frac{s}{s^2+25}\right\} + \frac{5}{13}\mathcal{L}^{-1}\left\{\frac{5}{s^2+25}\right\} + \frac{1}{13}\mathcal{L}^{-1}\left\{\frac{1}{(s-1)}\right\}$$

Finally:

$$\boxed{\boldsymbol{y = \frac{1}{13}e^t + \frac{5}{13}\sin(5t) - \frac{1}{13}\cos(5t)}}$$

39. $y'' + 5y' + 4y = 0 \mid y(0) = 1, y'(0) = 0$

Taking the Laplace of both sides gives us:

$$s^2Y(s) - sy(0) - y'(0) + 5(sY(s) - y(0)) + 4Y(s) = 0$$

Simplifying further:

$$s^2Y(s) - s + 5sY(s) - 5 + 4Y(s) = 0$$

Rearranging and factoring:

$$Y(s)(s^2 + 5s + 4) = 5 + s$$

Isolating $Y(s)$:

$$Y(s) = \frac{5+s}{(s+1)(s+4)}$$

Completing PFD:

$$\frac{5+s}{(s+1)(s+4)} = \frac{A}{(s+1)} + \frac{B}{(s+4)}$$

Getting a common denominator:

$$\frac{5+s}{(s+1)(s+4)} = \frac{As+4A+Bs+B}{(s+1)(s+4)}$$

Comparing coefficients:

$$A + B = 1$$

$$4A + B = 5$$

Matrix of system is: $\begin{pmatrix} 1 & 1 & 1 \\ -4 & -1 & 5 \end{pmatrix} \longrightarrow$ RREF $\begin{pmatrix} 1 & 0 & \frac{4}{3} \\ 0 & 1 & -\frac{1}{3} \end{pmatrix}$

Which gives us:

$$A = -\frac{4}{3}, B = -\frac{1}{3}$$

Thus our PFD result is:

$$\frac{4}{3(s+4)} - \frac{1}{3(s+1)}$$

Applying the inverse Laplace to both sides:

$$y = \mathcal{L}^{-1}\left\{\frac{4}{3(s+1)}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{3(s+4)}\right\}$$

Which gives us our final solution of:

$$\boldsymbol{[y = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t}]}$$

40. $y'' - 4y' = 6e^{3t} - 3e^{-t} \mid y(0) = 1, y'(0) = -1$

Apply $\mathcal{L}\{\}$ to both sides:

$$s^2Y(s) - sy(0) - y'(0) - 4sY(s) - 4y(0) = \frac{6}{s-3} - \frac{3}{s+1}$$

Substituting our initial conditions:

$$s^2Y(s) - s + 1 - 4sY(s) - 4 = \frac{6}{s-3} - \frac{3}{s+1}$$

Isolate $Y(s)$ and add over the -3 and $-s$:

$$Y(s)(s^2 - 4) = \frac{6}{s-3} - \frac{3}{s+1} + s + 3$$

Combining our first two fractions results in:

$$\frac{6s+6-3s+9}{(s-3)(s+1)} + s + 3 \longrightarrow \frac{3s+15}{(s-3)(s+1)} + s + 3$$

Adding the s and the 3:

$$\frac{3s+15+s^3-2s^2-3s+3s^2-6s-9}{(s-3)(s+1)} \longrightarrow \frac{s^3+s^2-6s+6}{(s-3)(s+1)}$$

Dividing over the denominator from $Y(s)$:

$$\frac{s^3+s^2-6s+6}{(s-3)(s+1)(s^2-4)}$$

Difference of perfect squares on the $(s^2 - 4)$:

$$\frac{s^3+s^2-6s+6}{(s-3)(s+1)(s-2)(s+2)}$$

Setting up the PFD of our fraction:

$$\frac{s^3+s^2-6s+6}{(s-3)(s+1)(s-2)(s+2)} = \frac{A}{s-3} + \frac{B}{s+1} + \frac{C}{s-2} + \frac{D}{s+2}$$

Addin each fraction:

$$\frac{As^3+Bs^3+Cs^3+Ds^3-3As^2+Bs^2-4Ds^2-4As-4Bs-7Cs+Ds+12A-4B-6C+6D}{(s-3)(s+1)(s-2)(s+2)}$$

The coefficient matrix for our numerator gives us:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -3 & 1 & 0 & -4 & 1 \\ -4 & -4 & -7 & 1 & -6 \\ 12 & -4 & -6 & -6 & 6 \end{pmatrix} \longrightarrow \text{RREF: } \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{3}{5} \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & -\frac{1}{10} \end{pmatrix}$$

Thus our corresponding coefficients are:

$$A = 0, B = \frac{3}{5}, C = \frac{1}{2}, D = -\frac{1}{10}$$

Thus our decomposed fraction results in:

$$Y(s) = \frac{3}{5(s+1)} + \frac{1}{2(s-2)} - \frac{1}{10(s+2)}$$

Taking $\mathcal{L}^{-1}$ of both sides gives us:

$$y = \frac{3}{5}e^{-t} + \frac{1}{2}e^{2t} - \frac{1}{10}e^{-2t}$$